Advanced Micro-Controller Systems

- Introduction, Microcontroller basics
- 32-bit Microcontroller technology
- ARM Cortex M Architecture, (CM0 CM3 CM4 CM7)
- Cortex M Microcontroller Instruction Set
- Exception Handling, Interrupts
- Real-time Operating Systems (RTOS)
- Memory Management, protected memory, MPU
- Implementing Calculations, Data Formats, FPU
- I/O, Device drivers
- Application Design
- Case Studies, Advanced Design Patterns

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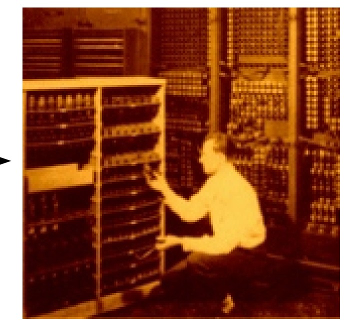
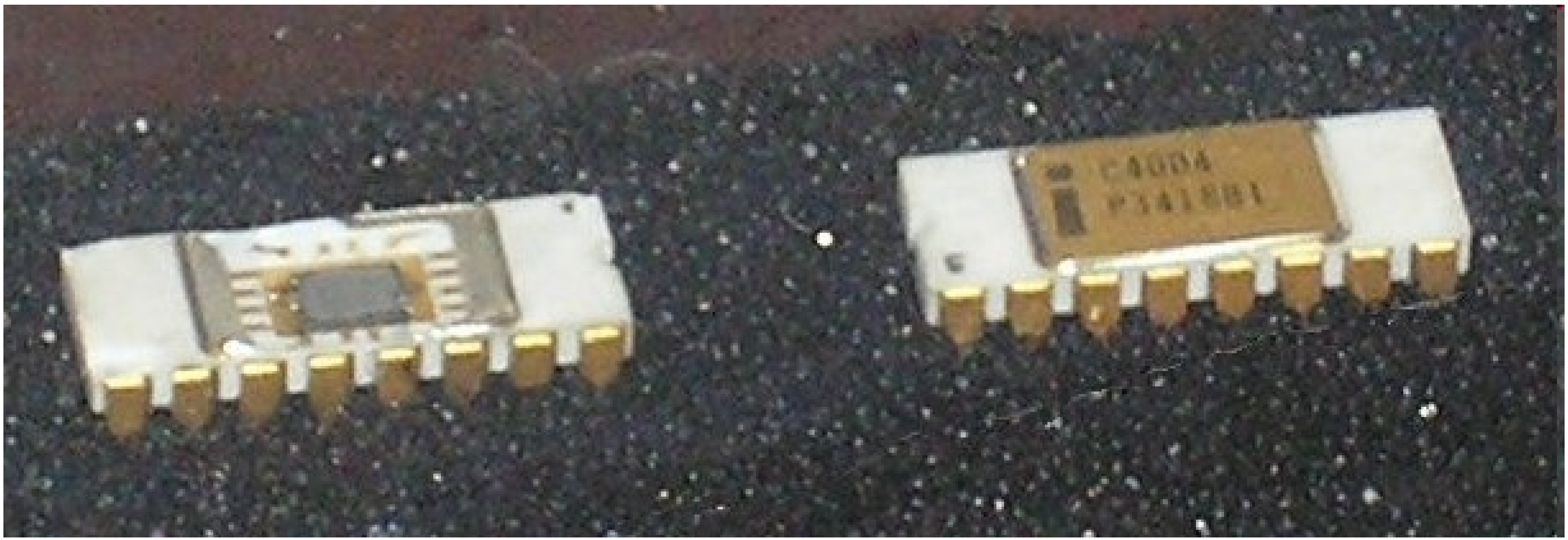
Computer Architecture Milestones

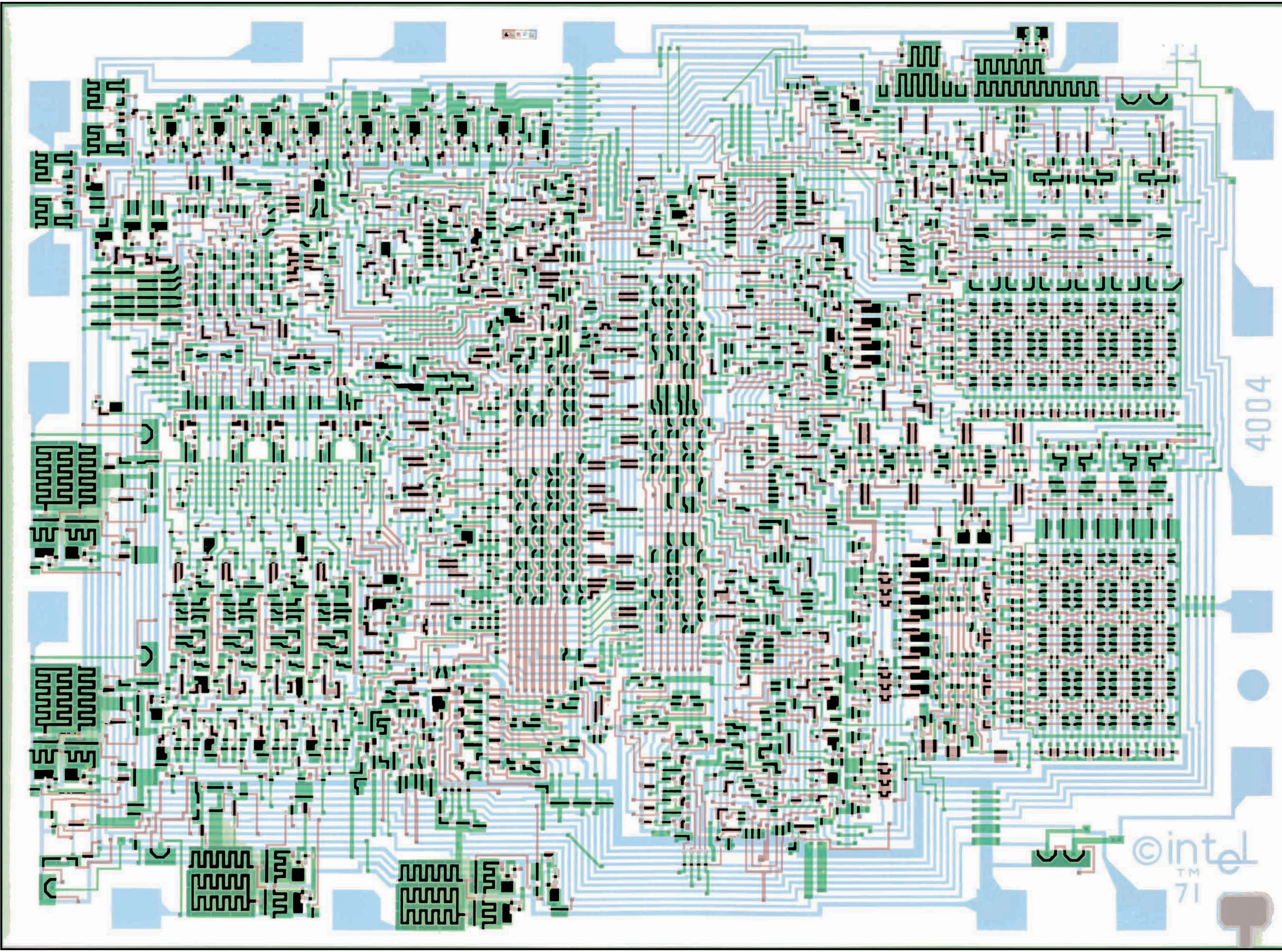
1834	Babbage	First try to build a digital computer
1936	C. Zuse	First scientific relay Computer
1943	Colossus	First electronic computer
1946	Eniac	First useful numerical computer
1951	Whirlwind	First real-time computer
1960	PDP1	First minicomputer (50 pcs sold)
1965	PDP8	Minicomputer (50.000 pcs sold)
1971	4004	First integrated microprocessor CPU
1981	IBM PC	Universal personal computer
1993	Newton	Palmtop computer



First Microprocessor

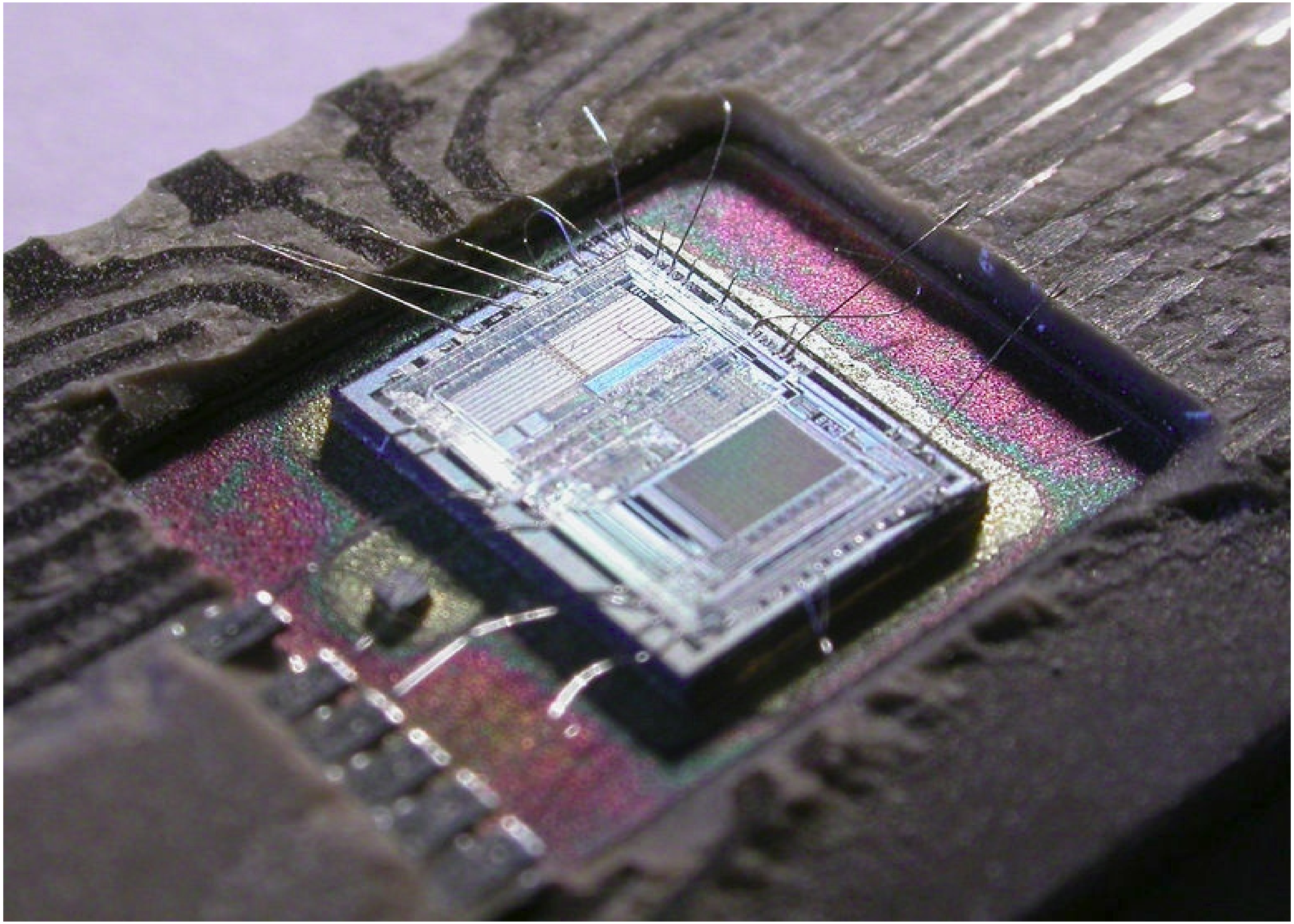


First 4 Bit Mikroprocessor: Intel 4004 (1971)


4004
©intel
TM
71

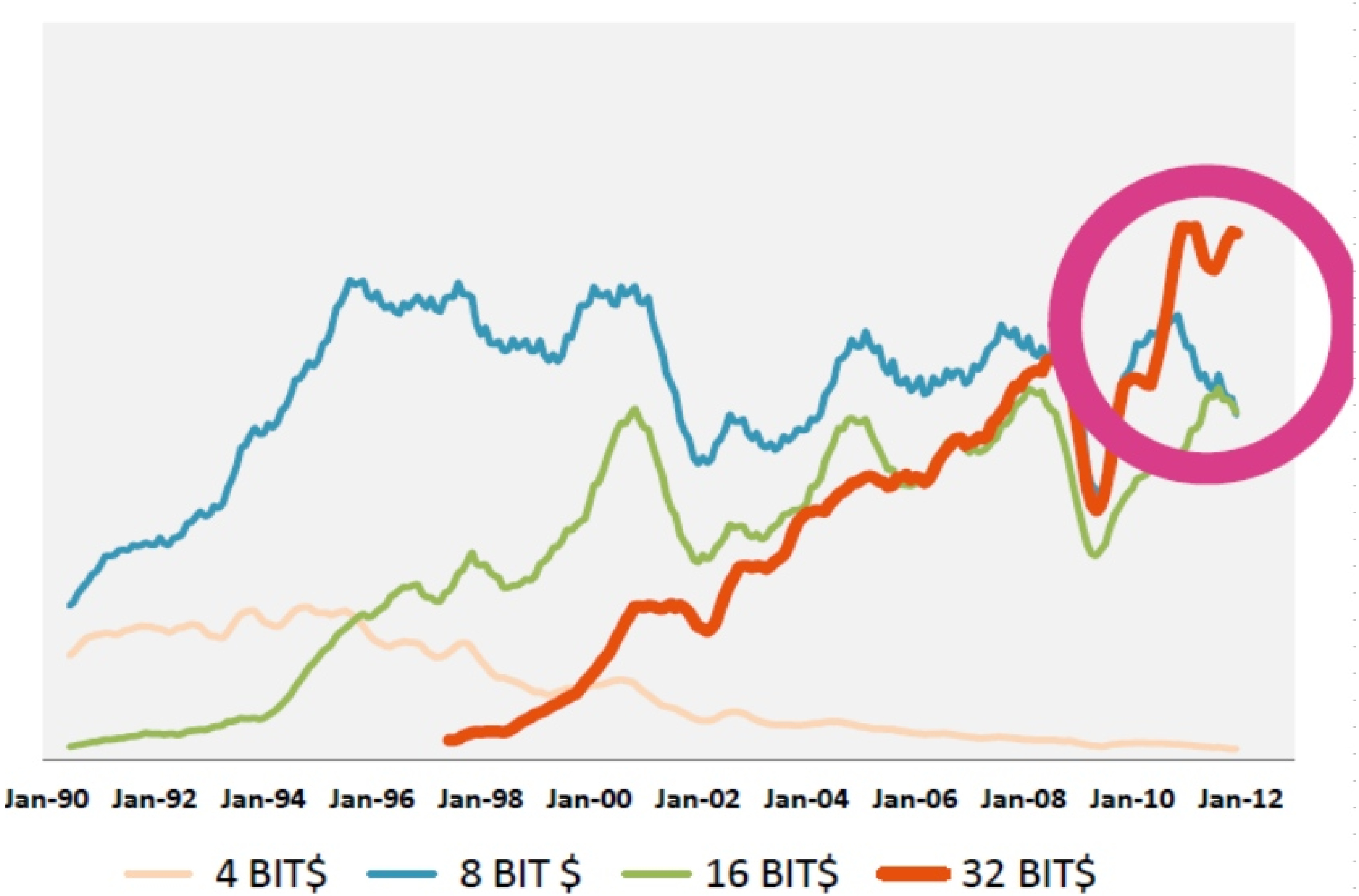
1976: First Microcontroller

INTEL 8742



Source: Wikipedia

CPU Architecture change



Source: WSTS (Worldwide Semiconductor Trade Statistics)

ARM Cortex M Manufacturers

Silicon labs

INTEL

Actel

Fujitsu

Toshiba

ATMEL/Microchip

Cypress Semiconductor

Freescale

NXP

Energy Micro

ST Microelectronics

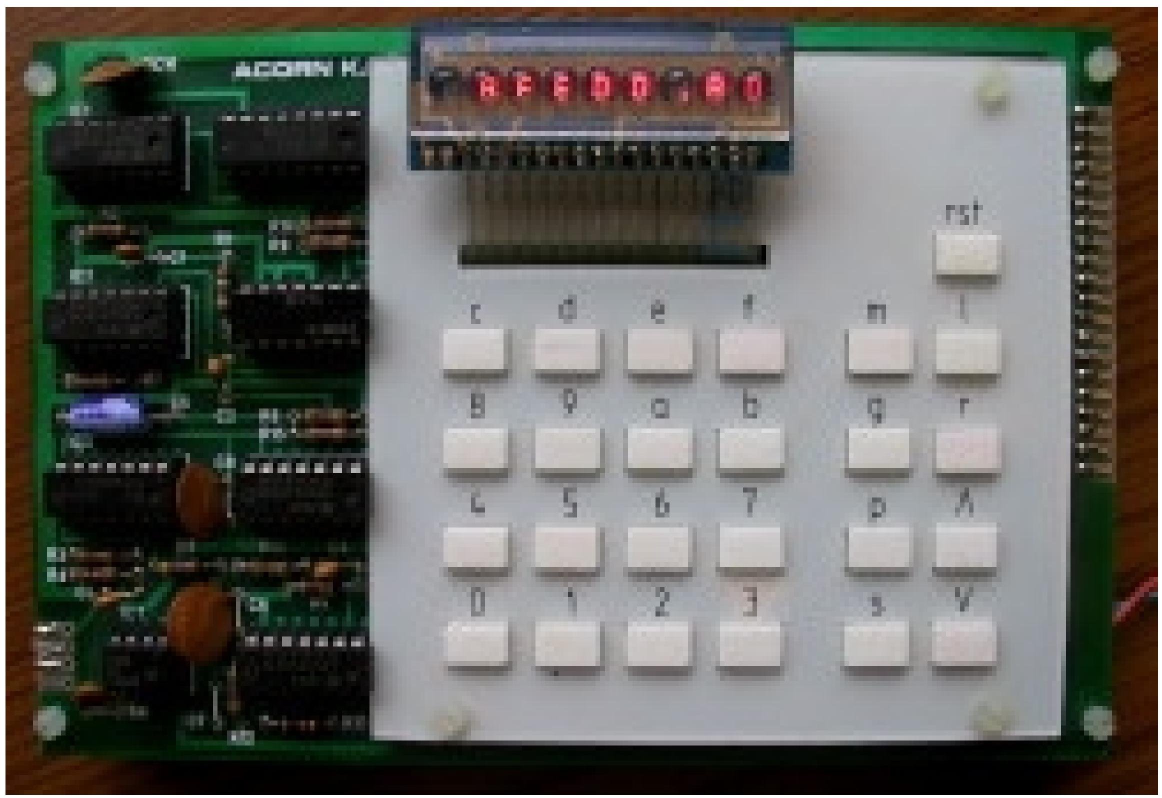
Texas Instruments

And probably many more...

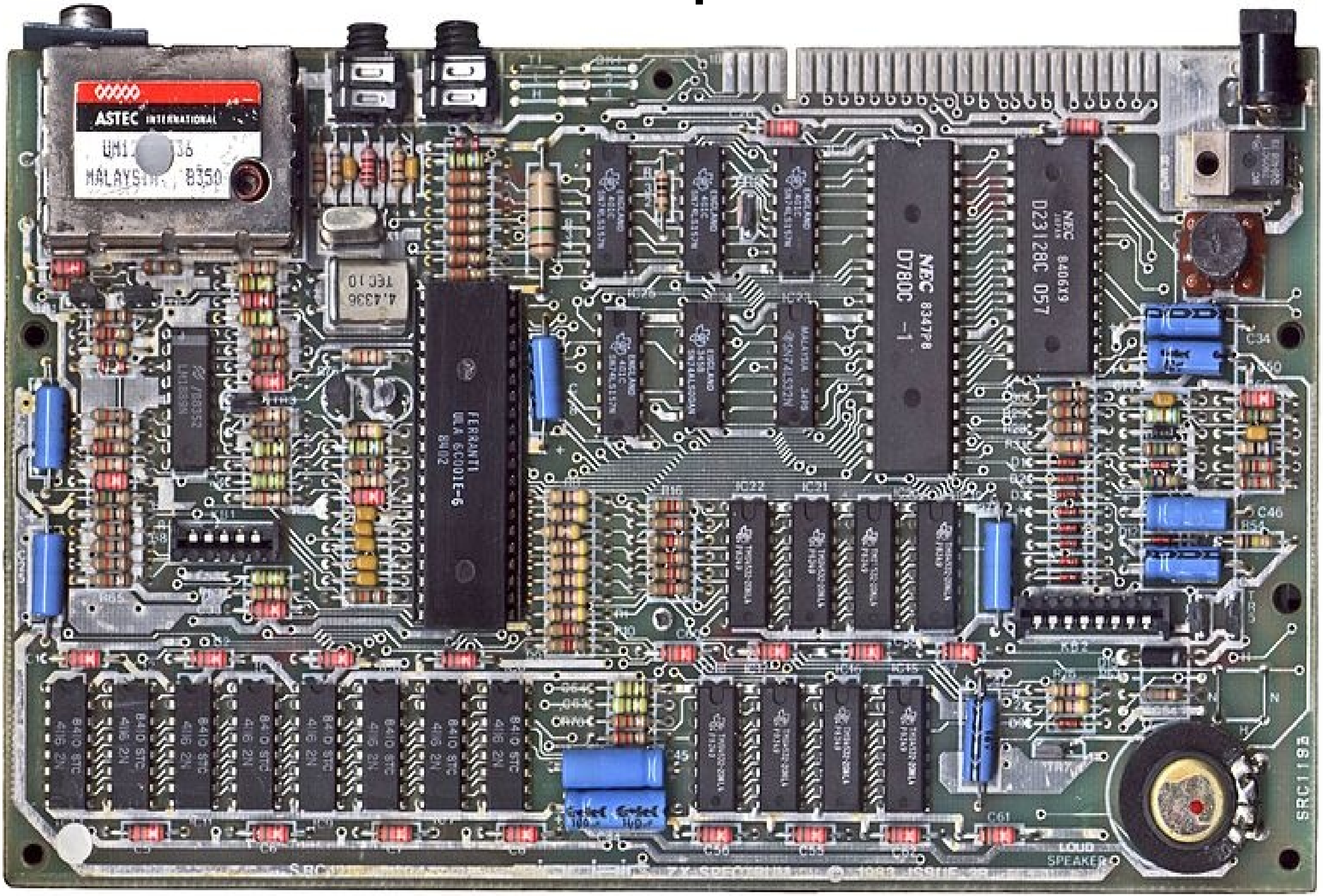
ARM Holding

Wikipedia.org

Acorn System 1



Sinclair ZXspectrum



Development
of the ARM Architecture

Acorn RISC Machine (1983-1985),
Rockwell 6502 Processor

http://www.ot1.com/arm/armchap1.htm
http://en.wikipedia.org/wiki/BBC_Micro

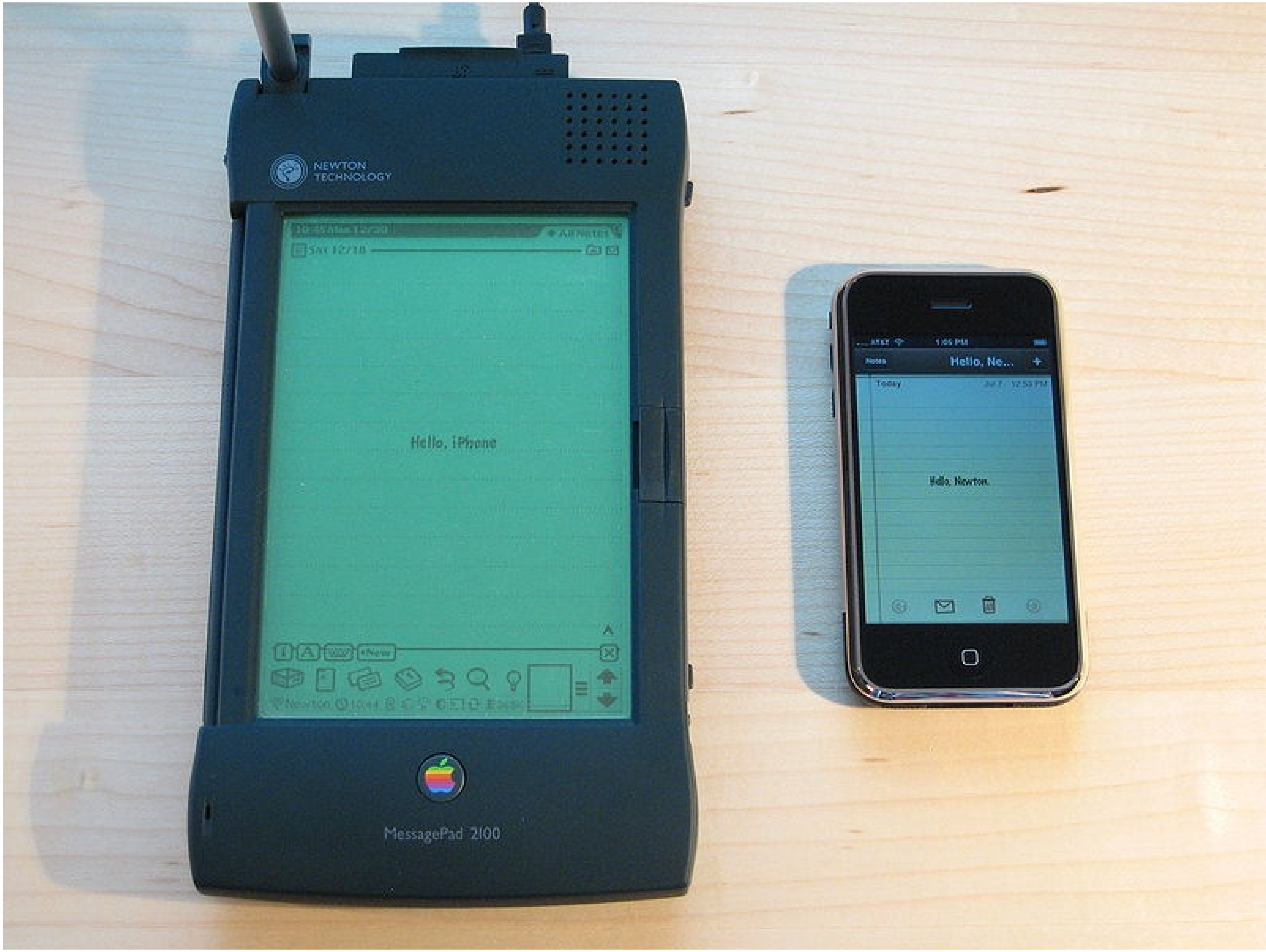
Acorn 500

ARM2

http://acorn.chriswhy.co.uk/Computers/A500.html

ARM2 on a FPGA 2009

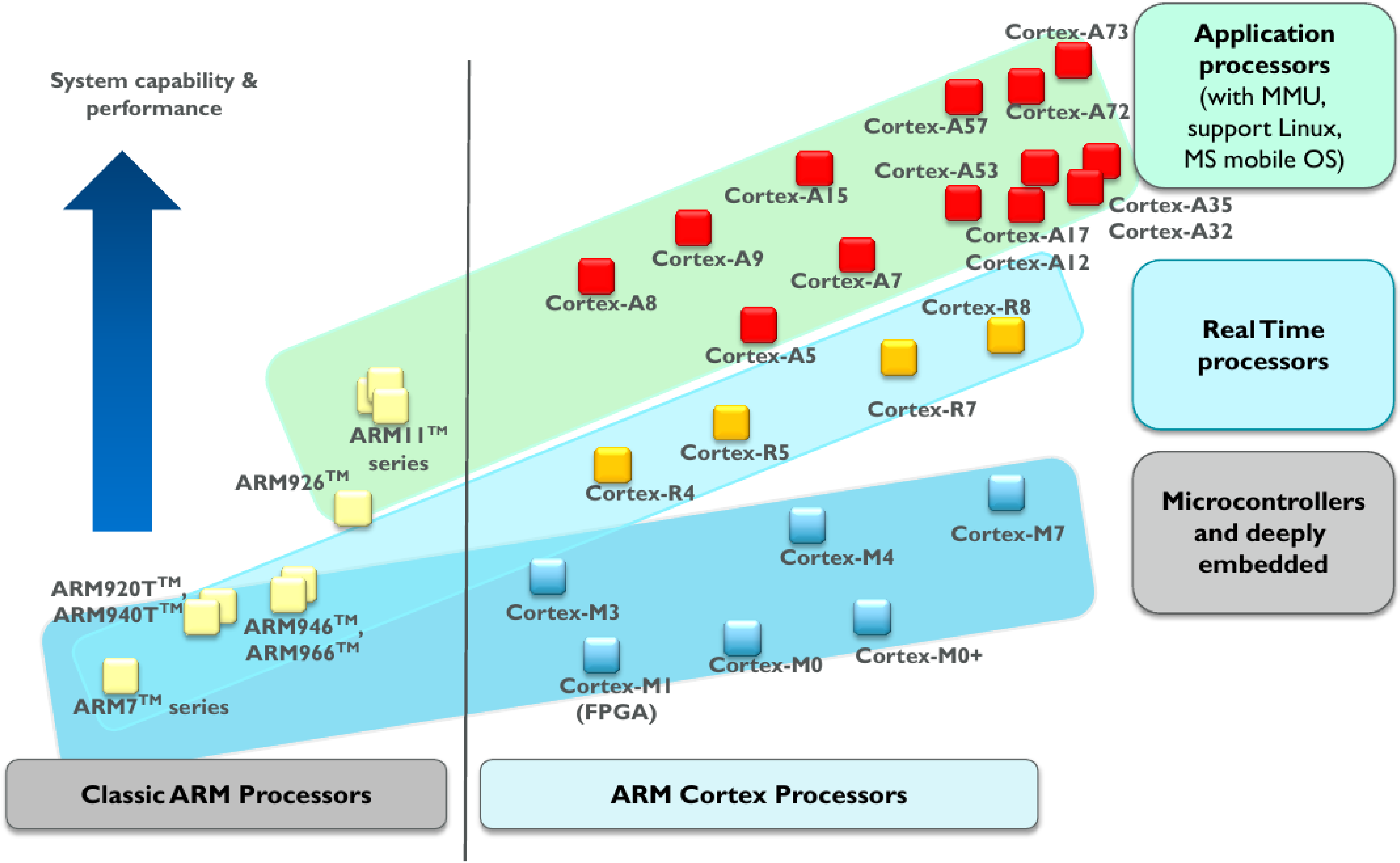
APPLE Newton / iPhone



http://upload.wikimedia.org/wikipedia/commons/7/76/Apple_Newton_and_iPhone.jpg

Steve Furber

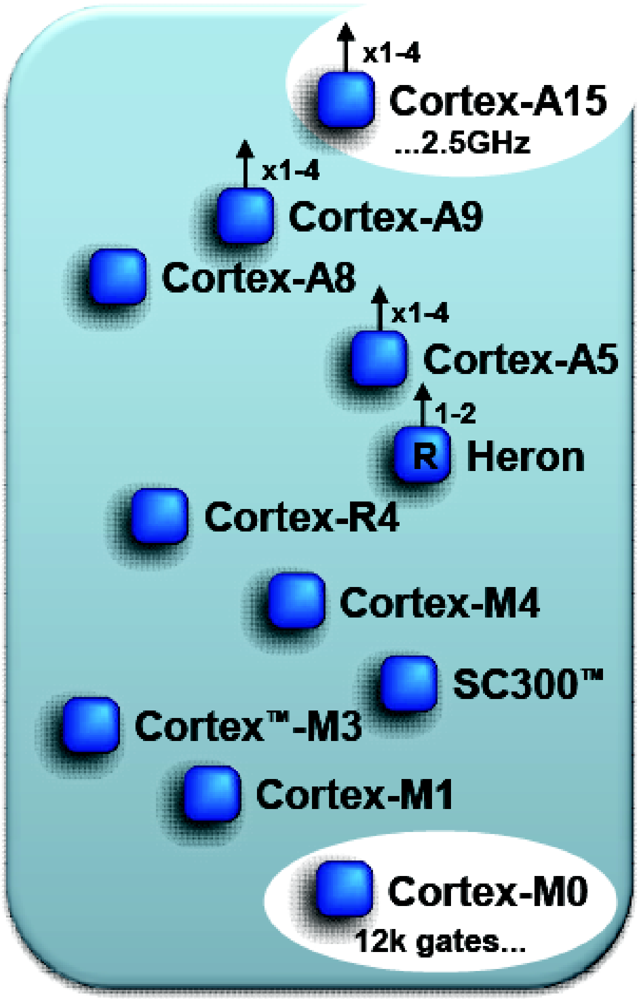
Professor Stephen Byram Furber CBE, FRS, FREng (born 1953 in Manchester, England) is the ICL Professor of Computer Engineering at the School of Computer Science at the University of Manchester but is probably best known for his work at Acorn where he was one of the designers of the BBC Micro and the ARM 32-bit RISC microprocessor.



System capability & performance
Cortex-A73
Cortex-A57
Cortex-A72
Cortex-A53
Cortex-A15
Cortex-A35
Cortex-A32
Cortex-A17
Cortex-A12
Cortex-A9
Cortex-A7
Cortex-A8
Cortex-A5
Cortex-R8
Cortex-R7
Cortex-R5
Cortex-R4
Cortex-M7
Cortex-M4
Cortex-M3
Cortex-M0
Cortex-M0+
Cortex-M1 (FPGA)
ARM11™ series
ARM926™
ARM920T™, ARM940T™
ARM946™, ARM966™
ARM7™ series
Application processors (with MMU, support Linux, MS mobile OS)
Real Time processors
Microcontrollers and deeply embedded
Classic ARM Processors
ARM Cortex Processors

ARM Cortex Processors (v7)

- ARM Cortex-**A** family (v7-A):
  - Applications processors for full OS and 3rd party applications
- ARM Cortex-**R** family (v7-R):
  - Embedded processors for real-time signal processing, control applications
- ARM Cortex-**M** family (v7-M):
  - Microcontroller-oriented processors for MCU and SoC applications

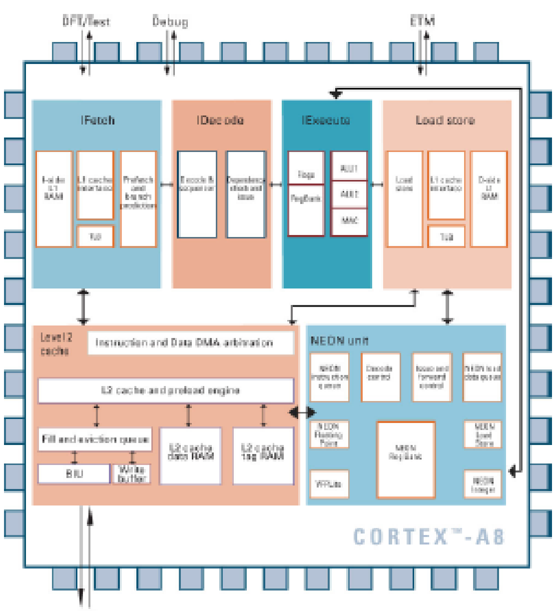


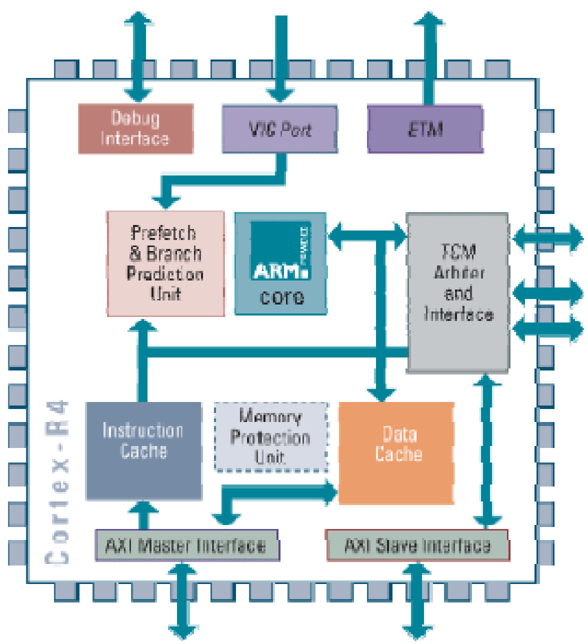
Cortex family

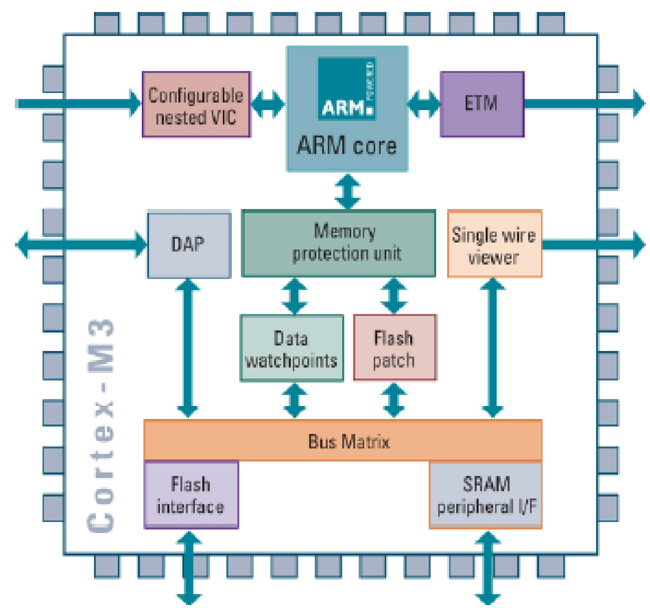
Cortex-A8
- Architecture v7A
- MMU
- AXI
- VFP & NEON support

Cortex-R4
- Architecture v7R
- MPU (optional)
- AXI
- Dual Issue

Cortex-M3
- Architecture v7M
- MPU (optional)
- AHB Lite & APB

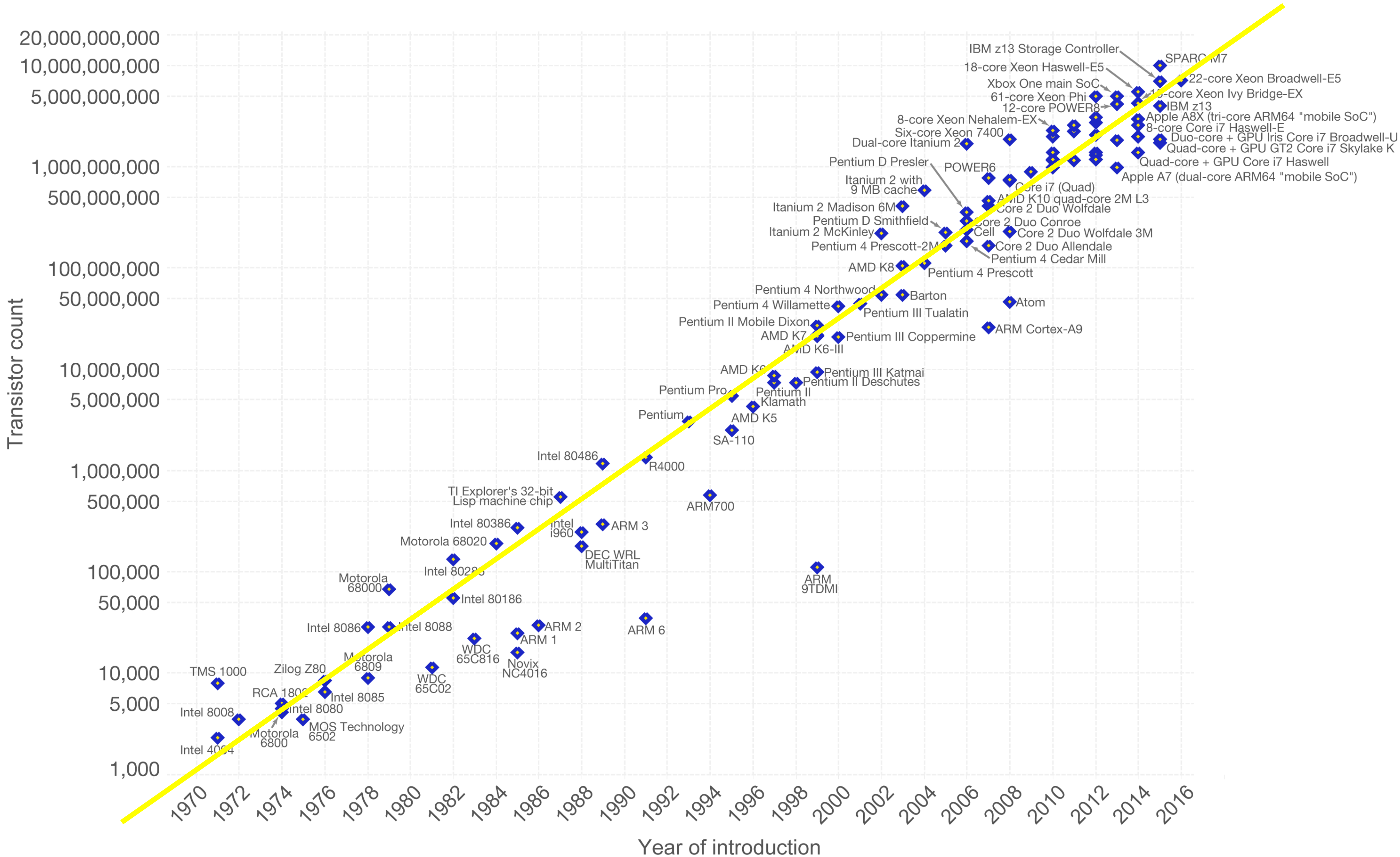






Moore's Law – The number of transistors on integrated circuit chips (1971-2016)

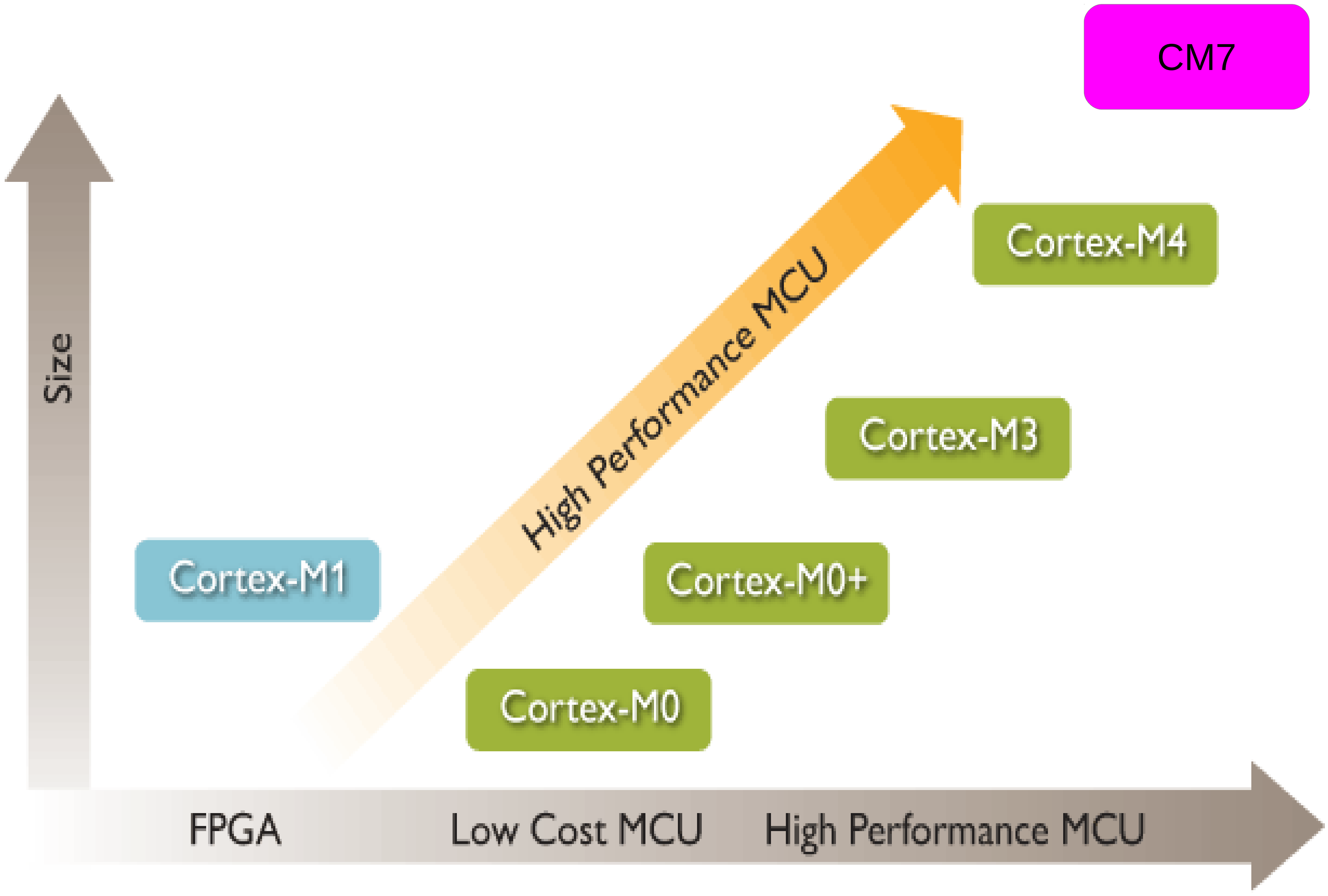
Moore's law describes the empirical regularity that the number of transistors on integrated circuits doubles approximately every two years. This advancement is important as other aspects of technological progress – such as processing speed or the price of electronic products – are strongly linked to Moore's law.



Data source: Wikipedia (https://en.wikipedia.org/wiki/Transistor_count)

The data visualization is available at OurWorldinData.org. There you find more visualizations and research on this topic.

ARM Cortex M Processor Family

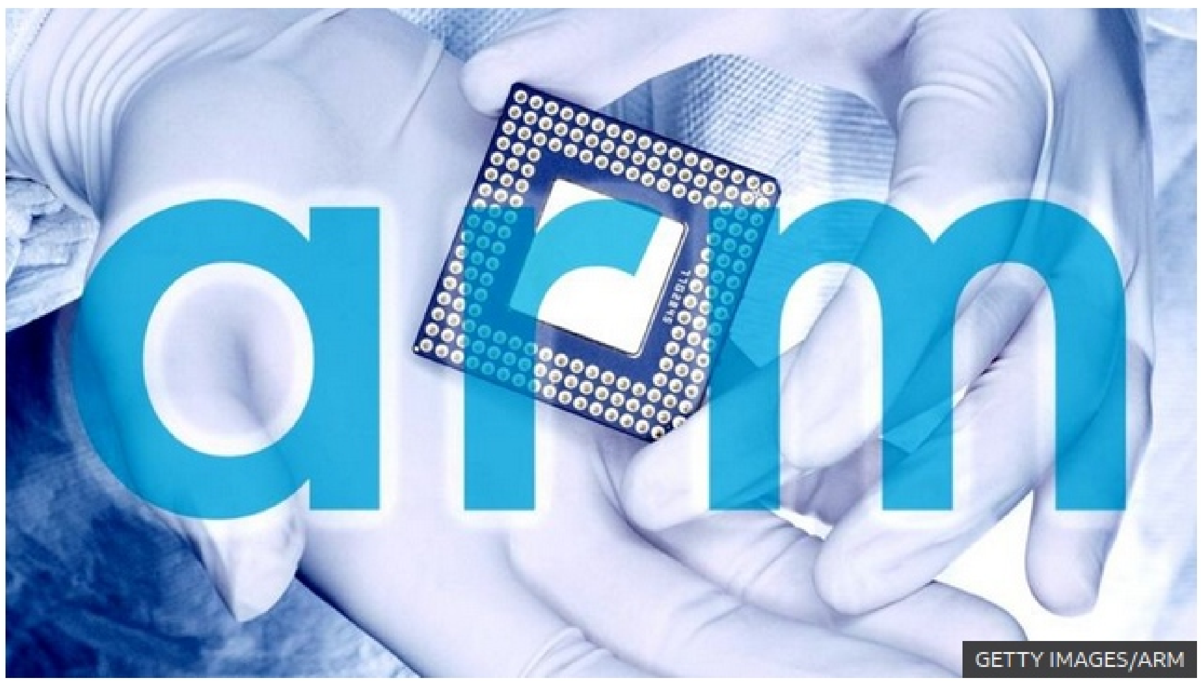


ARM: UK-based chip designer sold to US firm Nvidia

By Leo Kelion
Technology desk editor

14.9.2020

14 September



GETTY IMAGES/ARM

UK-based computer chip designer ARM Holdings is being sold to the American graphics chip specialist Nvidia.

The deal values ARM at $40bn (£31.2bn), four years after it was bought by Japanese conglomerate Softbank for $32bn.

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